Rapid multiplex PCR panel for pneumonia in hospitalised patients with suspected pneumonia in the USA: a single-centre, open-label, pragmatic, randomised controlled trial

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Summary

Background The clinical utility of rapid multiplex respiratory specimen PCR panels for pneumonia for patients with suspected pneumonia is undefined. We aimed to compare the effect of the BioFire FilmArray pneumonia panel (bioMérieux, Salt Lake City, UT, USA) with standard of care testing on antibiotic use in a real-world hospital setting.

Methods We conducted a single-centre, open-label, pragmatic, randomised controlled trial at the Mayo Clinic, Rochester, MN, USA. Hospitalised patients (aged ≥18 years) with suspected pneumonia, from whom expectorated or induced sputum, tracheal secretions, or bronchoalveolar lavage fluid respiratory culture samples (one per individual) could be collected during index hospitalisation, were eligible for inclusion. Samples from eligible participants were randomly assigned (1:1) with a computerised tool to undergo testing with either the BioFire FilmArray pneumonia panel, conventional culture, and antimicrobial susceptibility testing (intervention group) or conventional culture and antimicrobial susceptibility testing alone (control group). Antimicrobial stewardship review in both groups involved an assessment and recommendations for antibiotic modifications based on clinical data and the results from the BioFire FilmArray pneumonia panel, conventional culture, or both. The primary outcome was median time to first antibiotic modification (ie, escalation or de-escalation of antibiotics against Gram-negative and Gram-positive bacteria) within 96 h of randomisation, assessed with the Wilcoxon rank-sum test and analysed in a modified intention-to-treat population. This trial is registered with ClinicalTrials.gov (NCT05937126).

Findings Between Sept 15, 2020, and Sept 19, 2022, 1547 patients were screened for eligibility, of whom 1181 (76·3%) were randomly assigned: 582 (49·3%) to the intervention group and 599 (50·7%) to the control group. In total, 1152 participants were included in the modified intention-to-treat analysis, 589 (51·1%) in the control group and 563 (48·9%) in the intervention group. For the modified intention-to-treat population, median time to any first antibiotic modification was 20·4 h (95% CI 18·0–20·4) in the intervention group and 25·8 h (22·0–28·7) in the control group (p=0·076). Median time to any antibiotic escalation was 13·8 h (9·2–19·0) in the intervention group and 24·1 h (19·5–29·6) in the control group (p=0·0022). Median time to escalation of antibiotics against Gram-positive organisms was 10·3 h (6·2–30·9) in the intervention group and 24·6 h (19·5–37·2) in the control group (p=0·044); median time to escalation of antibiotics against Gram-negative organisms was 17·3 h (10·8–23·3) in the intervention group and 27·2 h (21·3–33·9) in the control group (p=0·010). Median time to any antibiotic de-escalation did not differ between groups (p=0·37). Median time to first de-escalation of antibiotics against Gram-positive organisms was 20·7 h (17·8–24·0) in the intervention group and 27·8 h (22·9–33·0) in the control group (p=0·015); median time to first de-escalation of antibiotics against Gram-negative organisms did not differ between groups (p=0·46).

Interpretation Clinical use of the BioFire FilmArray pneumonia panel might lead to faster antibiotic escalations, including for Gram-negative or Gram-positive bacteria, and faster antibiotic de-escalations directed at Gram-positive bacteria. Additional research is needed regarding antimicrobial de-escalation, especially when antibiotics with broad Gram-negative spectrum are being used, by use of rapid diagnostics in patients with lower respiratory tract infection.

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Introduction

Clinical practice guidelines for pneumonia recommend use of cultures to modify antibiotics.[1] However, culture-based methods are time-consuming and have low positivity rates.[1,2] Delays in pathogen-specific, appropriate antimicrobial therapy are associated with poor outcomes in

Research in context

Evidence before this study

Defining the microbiological cause of lower respiratory tract infections is challenging with conventional cultures. Antibiotics administered for suspected pneumonia are often empirically prescribed and might be unnecessarily broad-spectrum or not needed at all for some patients. New rapid molecular diagnostics might mitigate the inefficiencies of conventional cultures of respiratory or reveal pathogens not being treated. However, the effect of these rapid diagnostics on antibiotic use in the real world is not clear. We searched PubMed from the date of database inception to Dec 14, 2023, to identify research articles published in English using the following search terms: ("BIOFIRE or nucleic acid amplification or Unyvero or rapid diagnostic polymerase chain reaction (PCR)") AND ("pneumonia" or "community acquired pneumonia", or "hospital-acquired pneumonia" or "ventilator associated pneumonia") and ("antimicrobial or antibiotic stewardship" or "de-escalation" or "escalation" or "modification"). We found 228 studies, of which 52 were relevant. Previous studies showed that, although new nucleic acid amplification tests (NAATs), such as the BioFire FilmArray pneumonia panel (bioMérieux, Salt Lake City, UT, USA), provide rapid and accurate detection of common pneumonia pathogens and related antibiotic resistance determinants, only small studies suggest that use of respiratory NAATs might decrease inappropriate antibiotic use and duration. To date, no large, real-world studies have been done to assess the effect of these new NAAT diagnostics on antibiotic use in patients with suspected pneumonia.

Added value of this study

To our knowledge, this single-centre, open-label, pragmatic, randomised controlled trial is the first and largest study looking at the real-world effect of the BioFire FilmArray pneumonia panel on antibiotic use among patients with suspected pneumonia. Clinical use of this test could lead to faster escalations of antibiotics overall and faster de-escalations of antibiotics directed at Gram-positive bacteria, as well as provide more frequent and more timely opportunities for antimicrobial stewardship interventions.

Implications of all the evidence available

The available data show that rapid molecular diagnostics provide evidence-based opportunities for antimicrobial stewardship interventions. Based on these findings, rapid molecular diagnostics, such as the BioFire FilmArray pneumonia panel, have clinical utility in defining the microbiological causes of pneumonia and directing antimicrobial stewardship of antibiotics in pneumonia. However, further research is warranted in developing safe and reliable methods of antibiotic de-escalation directed at Gram-negative bacteria among patients with lower respiratory tract infection.

patients with lower respiratory tract infections. Additionally, difficulties in the interpretation of microbiological data can result in an overdiagnosis of pneumonia and inadequate de-escalation.[3,4] In an attempt to mitigate the deficiencies in conventional cultures and antimicrobial susceptibility testing (AST), faster and potentially more accurate nucleic acid amplification tests (NAATs),[5–10] such as the BioFire FilmArray pneumonia panel (bioMérieux, Salt Lake City, UT, USA), have been developed for the detection of common pathogens causing pneumonia with select antibiotic resistance determinants (eg, $bla_{\text{CTX-M}}$). NAATs performed on respiratory specimens have 96·2–98·4% positive and 96·8–98·1% negative agreement with conventional cultures and AST.[7,9–15]

Although small studies suggest that respiratory NAATs might decrease inappropriate antibiotic use and duration,[16–20] the clinical impact of these costly tests remains undefined. We aimed to assess the effect of a commercial multiplex PCR panel (BioFire FilmArray pneumonia panel) compared with conventional culture and AST (standard of care) on antibiotic use in patients suspected of lower respiratory tract infection in a real-world hospital setting. We hypothesised that the BioFire FilmArray pneumonia panel was associated with increased and faster identification of microorganisms causing lower respiratory tract infection with key antibiotic resistance determinants and, in conjunction with an antimicrobial stewardship review, would improve time to antibiotic modification.

Methods

Study design

We conducted a single-centre, open-label, pragmatic, randomised controlled trial at the Mayo Clinic, Rochester, MN, USA. Primary clinical care teams (PCCTs) made all decisions regarding patient management, including testing and antibiotics. An antimicrobial stewardship programme (ASP) member reviewed all participants.

The Mayo Clinic Institutional Review Board approved the study protocol. The institutional review board determined that the research involved no greater than minimal risk and approved a waiver of informed consent in accordance with 45 CFR 46.116, and a waiver of Health Insurance Portability and Accountability Act authorisation in accordance with applicable regulations. Additionally, given that the rapid test being studied was approved by the US Food and Drug Administration before the start of the study, its performance was not evaluated. To assess its effect on antibiotic use, the test needed to be run immediately with results reported promptly.

Participants

Adult (aged ≥18 years) patients with suspected pneumonia admitted to the Mayo Clinic Hospitals (Rochester, MN, USA), from whom respiratory samples (one per individual) meeting study enrolment criteria could be collected during index hospitalisation and from whom a respiratory culture was clinically ordered, were eligible for inclusion. Samples

meeting enrolment criteria comprised expectorated or induced sputum, tracheal secretions, or bronchoalveolar lavage fluid culture. Patients who were younger than 18 years, had specimens with poor quality Gram stain for expectorated sputum culture[21] or had positive respiratory cultures in the 7 days before the study (if known at randomisation), had previously enrolled in the study, had not provided Minnesota research authorisation, or were affected by the Global Data Protection Regulation were excluded.

PCCTs diagnosed pneumonia on the basis of clinical and radiographical findings. Pneumonia was defined as community-acquired if acquired in the community, hospital-acquired if onset was at least 48 h after hospital admission, or ventilator-associated if onset occurred at least 48 h after intubation. Other respiratory infections, such as tracheobronchitis, were characterised as other pulmonary infections. Patients with no clinical or laboratory evidence of infection were defined by PCCTs or ASP members as uninfected. All participant data, including baseline characteristics, information on hospitalisation or admission to the intensive care unit (ICU), microbiology results, and details of antibiotic therapy, were obtained from electronic health records (EHRs).

Randomisation and masking

Samples from eligible participants were randomly assigned (1:1) to undergo testing with either the BioFire FilmArray pneumonia panel plus conventional culture and AST (intervention group) or just conventional culture and AST (control group). Samples meeting inclusion criteria were randomised by laboratory technologists (APS and ADD) with a computerised tool (appendix p 4). Randomisation was stratified by specimen type for equal representation across the study groups. Only the first specimen collected from each study participant was randomised.

Study participants, ASP members, and PCCTs, including those assessing outcomes and analysing the data, were unmasked to group assignment given the visibility of BioFire FilmArray pneumonia panel results in the EHR.

Procedures

For both groups, standard laboratory methods were used and breakpoints and guidelines from the Clinical & Laboratory Standards Institute were followed for interpretation of susceptibilities (appendix p 26). Respiratory specimens were plated to sheep blood, chocolate, and eosin methylene blue agar (all three media were from Becton Dickinson, Franklin Lakes, NJ, USA and Remel, Lenexa, KS, USA) and incubated at 35°C for 2 days, with first examination after 16–24 h of incubation and results reported semi-quantitatively.

The BioFire FilmArray pneumonia panel used in the study did not include SARS-CoV-2 (appendix p 3) and was performed per package insert instructions 24 h per day 7 days a week, with results reported in the EHR. The viruses, *Legionella pneumophila, Chlamydia pneumoniae,* and *Mycoplasma pneumoniae* were reported as either present or absent, with other bacteria reported quantitatively (10^4 to $\geq 10^7$ copies per mL). Antibiotic guidance text based on clinical infectious diseases and microbiology consensus accompanied the BioFire FilmArray pneumonia panel test results in the EHR for specific pathogens (appendix pp 5–6). Other microbiological tests, such as serologies, urine antigen, blood cultures, and molecular tests (eg, influenza, respiratory syncytial virus, SARS-CoV-2, or nasal methicillin-resistant *Staphylococcus aureus* [MRSA]) were ordered at the discretion of PCCTs.

After participant specimen enrolment, the laboratory notified the study coordinators (LMH and KAR), who then notified the ASP team (appendix p 4). All new results were communicated via text to ASP pager and email to the ASP team from Monday to Friday between 0800 h and 1700 h. Participants who were enrolled and results that were reported outside of these hours, on weekends, or during holiday periods were communicated to the ASP team the following weekday morning. The ASP team consisted of nine physicians and two pharmacists rotating between members on a weekly basis. ASP conducted a review of all relevant clinical data (including clinical, microbiological, and BioFire FilmArray results) in EHRs and provided recommendations (including antibiotic modifications), as necessary, to PCCTs the duration of each participant's enrolment. Scenario-based standardised recommendations were developed and shared with the ASP team (appendix pp 9–15). The decision to intervene, provide specific recommendations, or both were at the discretion of the assigned ASP member. PCCTs made final decisions for specific antibiotic therapy, irrespective of ASP recommendations. In both study groups, ordering clinicians were notified when a clinically significant pathogen was identified as per institutional policy.

See Online for appendix

Outcomes

The primary outcome was mean time to first antibiotic modification (ie, escalation or de-escalation, including for antibiotics against Gram-negative and Gram-positive bacteria) within 96 h of randomisation. However, given that data were found not to be normally distributed, median (IQR) values are reported instead of mean (SD) values.

Time to escalation was calculated as the time from randomisation to the time that the escalated antibiotic was first administered, whereas time to de-escalation was the time from randomisation to the time that the last broad-spectrum antibiotic was administered. Antibiotic escalation and de-escalation were based on antibiotic activity (appendix pp 16–18). Antibiotics directed primarily at MRSA were considered to be antibiotics against Gram-positive bacteria. Antibiotics with a spectrum including Gram-negative bacteria were considered to be antibiotics against Gram-negative bacteria.

Secondary outcomes within 30 days of randomisation were mortality; hospital length of stay for participants alive at 30 days; length of stay in the ICU; ventilator-free days (ie, days without mechanical intubated ventilation, calculated by the date of the last day of mechanical

ventilation minus the date of randomisation); duration of broad-spectrum antibiotic therapy within 96 h of randomisation; mean time to antibiotic escalation or de-escalation within 96 h of randomisation (reported as median values because data were not normally distributed), further defined by Gram-positive coverage, Gram-negative coverage, or both; acquisition of new multidrug-resistant organisms with and without hospital-acquired infection; and hospital-onset infection with *Clostridioides difficile*.

The frequencies of ASP interventions, which included determination of the patient's clinical syndrome; regimen appropriateness at the start and end of the study period; documentation of time to first antibiotic modification, even if not advised by the ASP team; time to recommendations to PCCTs from enrolment (0–24, 24–48, and 48–96 h); and specific recommendations given, were measured. Specific recommendations were intravenous to oral conversion, de-escalation of therapy, escalation of therapy (starting at least one or more antibiotics or escalation of antibiotic spectrum), dose optimisation, therapeutic drug monitoring, shortening or defining duration of antibiotic therapy, infectious diseases consultation, or other.

We performed a post-hoc analysis to assess the concordance and time to positivity of on-panel organism identification and AST between the BioFire FilmArray pneumonia panel and conventional culture plus AST. Additional post-hoc analyses were performed of the median (IQR) time to first antibiotic modification for participants with no pathogen detected in either group by any method; participants with a single positive detection of *S aureus, Pseudomonas aeruginosa, Streptococcus pneumoniae, Escherichia coli*, or *Serratia marcescens* by culture in the control group or by the BioFire FilmArray pneumonia panel in the intervention group; participants with detection of *Streptococcus agalactiae, Moraxella catarrhalis*, or *Haemophilus influenzae* by the BioFire FilmArray pneumonia panel in the intervention group compared with detection of usual flora among participants in the control group; participants with SARS-CoV-2 detected in either group; and uninfected participants in both groups. In post-hoc analyses, we also evaluated the effect of the BioFire FilmArray pneumonia panel on time to first antibiotic modification among participants in the intervention group by specimen type (ie, expectorated sputum, tracheal secretions, and bronchoalveolar lavage fluid); ICU admission; 30-day re-admission; antibiotic intensity score within the first 96 h following randomisation; and aggregate standardised costs (appendix p 28). The primary endpoint analyses comparing the proportions of participants who were versus who were not admitted to ICU at baseline and the proportions of patients with neutropenia versus without at baseline will be explored separately. The primary endpoint analysis excluding the first hour after randomisation to account for changes related to Gram stain results was not completed because the time for Gram stain results was not collected.

Statistical analysis

This study was designed to detect a difference in the primary outcome (ie, mean time to first antibiotic modification in the first 96 h following randomisation). 1200 participants (600 in each group) were planned for inclusion to yield at least 200 evaluable participants with a positive detection in each study group. Approximately 40% of participants were anticipated to have positive cultures (20% for on-panel organisms) and 30% to have a positive BioFire FilmArray pneumonia panel result. This sample size was expected to provide 80% power with a two-sided alpha level of 0·05 to detect a difference of 0·28 SD or more in the time to first antibiotic modification between the two groups. This size translates to a detectable difference of at least 9 h (SD 32 h). With 1200 participants, if the SD was larger than 32 h, there would still be sufficient power to detect a difference between groups of at least 10 h.

The modified intention-to-treat population was used to evaluate the effect of the BioFire FilmArray pneumonia panel and included participants whose specimen met all eligibility criteria and no exclusion criteria. Cohorts of participants with community-acquired, hospital-acquired, or ventilator-associated pneumonia (ie, the pneumonia cohort) were analysed separately.

We calculated the median differences (95% CIs) in time to first antibiotic modification within 96 h following randomisation in both groups. For all study participants, including those who died within 96 h, time of earliest antibiotic modification was used as the time to first antibiotic modification. Participants with no antibiotic modification were assigned a time of 96 h. The Wilcoxon rank-sum test (used because data were not normally distributed) was the primary test used to assess whether the median values of the primary outcome were different between groups. Among the specimens randomly assigned to the intervention group, comparisons of proportions of specific microorganisms detected (ie, paired proportions) were performed with McNemar's tests. In the subset of participants who had a microorganism detected, we performed comparisons of time to positive microbiology (in h) using Wilcoxon signed rank tests. We estimated the probability of escalation or de-escalation of antibiotic use between 0 h and 96 h following randomisation using the empirical cumulative distribution function with pointwise 95% CIs. All tests were two-sided and p values of less than 0·05 were considered statistically significant.

We performed all analyses using SAS (version 9.4). This trial is registered with ClinicalTrials.gov (NCT05937126).

Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

Between Sept 15, 2020, and Sept 19, 2022, 1547 patients were screened for eligibility, from whom 1181 (76·3%) specimens were randomly assigned: 582 (49·3%) to the

intervention group and 599 (50·7%) to the control group (figure 1). In total, 1152 participants were included in the modified intention-to-treat analysis, 589 (51·1%) in the control group and 563 (48·9%) in the intervention group. Among all 1152 participants at randomisation, median age was 64 years (IQR 54·0–74·0), 733 (63·6%) participants were men, 995 (86·4%) participants were White, and 727 (63·1%) participants had already been admitted to the ICU (table 1). The median Charlson Comorbidity Index was 4 (IQR 2–6). 514 (44·6%) participants were characterised as having pneumonia and were included in the pneumonia cohort: 234 (45·5%) had community-acquired pneumonia, 181 (35·2%) had hospital-acquired pneumonia, and 99 (19·3%) had ventilator-associated pneumonia. Among participants without community-acquired, hospital-acquired, or ventilator-associated pneumonia, diagnoses included non-pulmonary infection, acute exacerbation of chronic obstructive pulmonary disease, aspiration, atelectasis, empyema, lung abscess, other pulmonary infections (eg, fungal pneumonia), mucus plug, non-infectious pulmonary process, pulmonary oedema, COVID-19, and other viral respiratory tract infections.

Respiratory sample types were equally distributed between groups (appendix p 29). Among the 514 participants in the pneumonia cohort, respiratory samples comprised sputum (87 [34·3%] of 254 participants in the intervention group *vs* 89 [34·2%] of 260 participants in the control group), induced sputum (15 [5·9%] *vs* 12 [4·6%]), tracheal secretions (101 [39·8%] *vs* 84 [32·3%]), and bronchoalveolar lavage (51 [20·1%] *vs* 75 [28·8%]). Conventional culture showed no growth of microorganisms in 83 (14·7%) of 563 participants in the intervention group and 75 (12·7%) of 589 participants in the control group. The top five microorganisms identified among participants in the control group (listed in order of most to least detected) were *S aureus, P aeruginosa, E coli, S marcescens*, and *S pneumoniae*. In the intervention group, the top five microorganisms identified were *S aureus, H influenzae, P aeruginosa, S agalactiae,* and *E coli* by the BioFire FilmArray pneumonia panel and *S aureus, P aeruginosa, E coli, Klebsiella pneumoniae* group, and *S marcescens* by conventional culture. 252 (44·8%) of 563 participants tested with the BioFire FilmArray pneumonia panel had at least one microorganism detected, with 389 detections in total (table 2).

Within the concordant specimens of the intervention group, median time to microbial detection was 2·0 h (IQR 2·0–2·2) with the BioFire FilmArray pneumonia panel versus 31·3 h (29·2–36·6) with conventional culture (table 2). The panel identified significantly higher proportions of *S agalactiae, S pneumoniae, M catarrhalis*, and *H influenzae* than did conventional culture (table 2). The panel identified 14 determinants of resistance in Gram-negative bacteria in samples from 11 participants (11 $bla_{\text{CTX-M}}$, two bla_{KPC}, and one $bla_{\text{OXA-48 like}}$). Three (27·3%) of 11 $bla_{\text{CTX-M}}$ detections were discordant between the panel and conventional culture plus AST (two were reported as susceptible to ceftriaxone by AST, one had no growth on conventional culture). Carbapenem resistance and bla_{KPC} detection were 100%

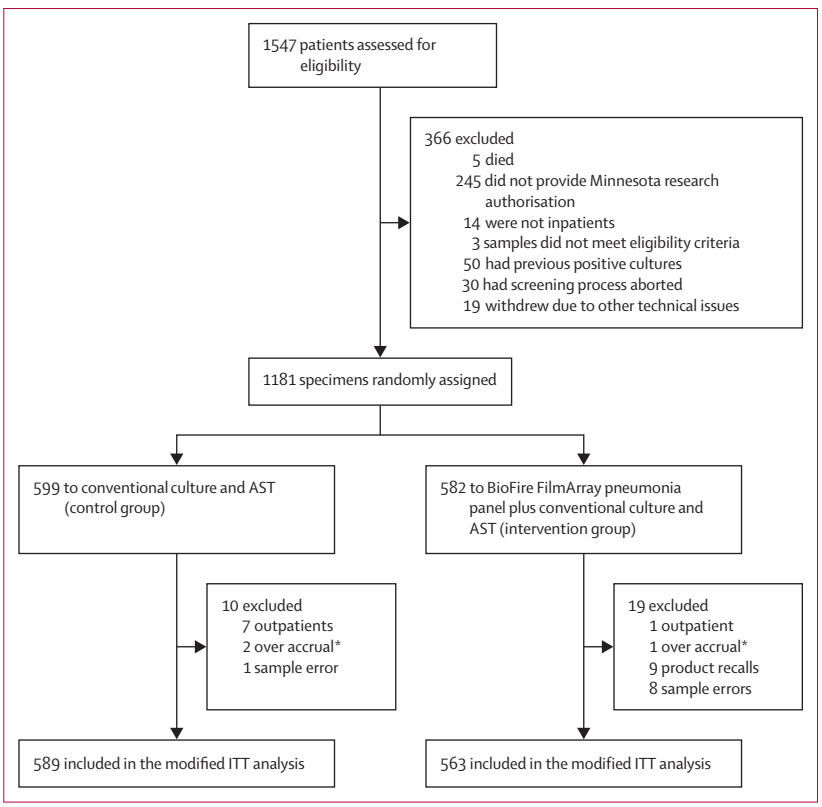


Figure 1: **Trial profile**
The modified ITT population included all participants whose specimen met all eligibility criteria and no exclusion criteria and was used to evaluate test performance. AST=antimicrobial susceptibility testing. ITT=intention-to-treat. *Over accrual refers to specimens that were removed due to being over the Institutional Review Board's approved limit.

concordant. The panel detected *Legionella pneumophila* in five participants, with no detection of *M pneumoniae* or *C pneumoniae*.

The BioFire FilmArray pneumonia panel identified 42 (7·5%) non-SARS-CoV-2 viruses in 563 specimens, including one (0·18%) adenovirus, four (0·71%) non-SARS-CoV-2 coronavirus, one (0·18%) influenza A, two (0·35%) human metapneumovirus, five (0·88%) parainfluenza virus, 26 (4·62%) human rhinovirus or enterovirus, and three (0·53%) respiratory syncytial virus. By contrast, in the control group, independent viral testing obtained for 189 participants at the discretion of the PCCTs was positive for influenza virus or respiratory syncytial virus in five (2·6%) of 189 participants. Standalone SARS-CoV-2 testing was positive in 45 (11·9%) of 379 participants tested in the intervention group and in 35 (8·9%) of 395 participants in the control group. The most common off-panel microorganisms detected by conventional culture across both study groups were *Stenotrophomonas maltophilia, Citrobacter koseri,* and *Morganella* species.

Blood cultures were positive in 18 (3·8%) of 480 participants in the intervention group and 23 (4·8%) of 483 participants in the control group. Nasal MRSA screening was performed on 575 participants (273 in the intervention group and 302 in the control group), with

	Modified ITT cohort			Pneumonia cohort		
	Intervention group (n=563)	Control group (n=589)	Total (n=1152)	Intervention group n=254)	Control group (n=260)	Total (n=514)
Sex						
Female	213 (37·8%)	206 (35·0%)	419 (36·4%)	93 (36·6%)	87 (33·5%)	180 (35·0%)
Male	350 (62·2%)	383 (65·0%)	733 (63·6%)	161 (63·4%)	173 (66·5%)	334 (65·0%)
Race						
White	480 (85·3%)	515 (87·4%)	995 (86·4%)	216 (85·0%)	231 (88·8%)	447 (87·0%)
Black	19 (3·4%)	15 (2·5%)	34 (3·0%)	8 (3·1%)	6 (2·3%)	14 (2·7%)
Asian	6 (1·1%)	9 (1·5%)	15 (1·3%)	5 (2·0%)	4 (1·5%)	9 (1·8%)
Native American or Alaska Native	7 (1·2%)	6 (1·0%)	13 (1·1%)	4 (1·6%)	4 (1·5%)	8 (1·6%)
Other	30 (5·3%)	28 (4·8%)	58 (5·0%)	12 (4·7%)	8 (3·1%)	20 (3·9%)
Unknown	21 (3·7%)	16 (2·7%)	37 (3·2%)	9 (3·5%)	7 (2·6%)	16 (3·1%)
Age, years	62·4 (53·0–74·0)	62·2 (54·0–73·0)	62·3 (54·0–74·0)	64·0 (54·0–74·0)	65·5 (56·5–72·0)	65·0 (55·0–73·0)
Length of time from admission to randomisation, days	2 (1–6)	2 (1–7)	2 (1–7)	3 (1–8)	2 (1–7)	2 (1–7)
Coexisting conditions						
Charlson Comorbidity Index	4·1 (2·0–6·0)	4·2 (2·0–6·0)	4·2 (2·0–6·0)	4·0 (2·0–6·0)	4·0 (3·0–6·0)	4·0 (2·0–6·0)
Diabetes with or without end-organ damage	152 (27·0%)	179 (30·4%)	331 (28·7%)	75 (29·5%)	85 (32·7%)	160 (31·1%)
Myocardial infarction	53 (9·4%)	76 (12·9%)	129 (11·2%)	26 (10·2%)	36 (13·8%)	62 (12·1%)
Congestive heart failure	123 (21·8%)	124 (21·1%)	247 (21·4%)	66 (26·0%)	61 (23·5%)	127 (24·7%)
Dementia	4 (0·7%)	2 (0·3%)	6 (0·5%)	2 (0·8%)	1 (0·4%)	3 (0·6%)
Renal disease (creatinine >3 mg/dL, renal transplant, uremia)	111 (19·7%)	134 (22·8%)	245 (21·3%)	50 (19·7%)	65 (25·0%)	115 (22·4%)
Chronic obstructive pulmonary disease	89 (15·8%)	105 (17·8%)	194 (16·8%)	43 (16·9%)	53 (20·4%)	96 (18·7%)
Any solid tumour in previous 5 years with or without metastasis	105 (18·7%)	98 (16·6%)	203 (17·6%)	42 (16·5%)	46 (17·7%)	88 (17·1%)
Leukaemia or lymphoma	64 (11·4%)	74 (12·6%)	138 (12·0%)	23 (9·1%)	34 (13·1%)	57 (11·1%)
Any liver disease (mild, moderate, or severe)	22 (3·9%)	19 (3·2%)	41 (3·6%)	11 (4·3%)	11 (4·2%)	22 (4·3%)
HIV or AIDS	3 (0·5%)	2 (0·3%)	5 (0·4%)	1 (0·4%)	2 (0·8%)	3 (0·6%)
Solid organ transplant	35 (6·2%)	48 (8·1%)	83 (7·2%)	15 (5·9%)	19 (7·3%)	34 (6·6%)
Bone marrow transplant	20 (3·6%)	27 (4·6%)	47 (4·1%)	2 (0·8%)	11 (4·2%)	13 (2·5%)
Clinical status						
In ICU	357 (63·4%)	370 (62·8%)	727 (63·1%)	175 (68·9%)	173 (66·5%)	348 (67·7%)
Receiving ventilation	261 (46·4%)	292 (49·6%)	553 (48·0%)	129 (50·8%)	141 (54·2%)	270 (52·5%)
Neutropenia	34 (6·0%)	44 (7·5%)	78 (6·8%)	11 (4·3%)	15 (5·8%)	26 (5·1%)
Any pneumonia	254 (45·1%)	260 (44·1%)	514 (44·6%)	..	..	..
Community-acquired pneumonia	106 (18·8%)	128 (21·7%)	234 (20·3%)	106 (41·7%)	128 (49·2%)	234 (45·5%)
Hospital-acquired pneumonia	98 (17·4%)	83 (14·1%)	181 (15·7%)	98 (38·6%)	83 (31·9%)	181 (35·2%)
Ventilator-associated pneumonia	50 (8·9%)	49 (8·3%)	99 (8·6%)	50 (19·7%)	49 (18·8%)	99 (19·3%)
Severe pneumonia	NA	NA	NA	178 (70·1%)	184 (70·8%)	362 (70·4%)
Lung abscess	4 (0·7%)	6 (1·0%)	10 (0·9%)	NA	NA	NA
Tracheobronchitis	10 (1·8%)	5 (0·8%)	15 (1·3%)	NA	NA	NA

Data are n (%) or median (IQR). The modified ITT population included all participants whose specimen met all eligibility criteria and no exclusion criteria. The pneumonia cohort included all participants with community-acquired, hospital-acquired, or ventilator-associated pneumonia. ICU=intensive care unit. NA=not applicable. ITT=intention-to-treat.

Table 1: **Demographic and clinical characteristics of participants at randomisation**

no significant difference in negative nasal PCR rates between both groups (p=0·19). Among detections of *S aureus* with the BioFire FilmArray pneumonia panel, 18 detections tested positive for *mecA*. Results from the nasal MRSA PCR assay were concordantly positive in five detections, negative in five, and not performed in eight. Two participants tested positive with nasal MRSA screening; however, *S aureus* was not detected in their samples by the BioFire FilmArray pneumonia panel or conventional culture.

In the modified intention-to-treat analysis, 519 (92·2%) of 563 participants in the intervention group and 538 (91·3%) of 589 participants in the control group were prescribed at least one antibiotic within 96 h of randomisation, most commonly vancomycin, cefepime, piperacillin–tazobactam, or a combination of all three (table 3). Meropenem use was lower in the intervention than the control group (p=0·040; table 3). Overall, antibiotic spectrum at the start and end of the 96 h time period was similar between the cohort including all participants and the pneumonia cohort (data not shown).

	Control group (n=589)	Intervention group (n=563)				Time to positive microbiology among concordant samples in intervention group, h			
	Conventional culture	Conventional culture	BioFire FilmArray pneumonia panel	Concordance pathogen between cultures and panel (%)	p value*	Number of specimens	Conventional culture	BioFire FilmArray pneumonia panel	p value†
Acinetobacter calcoaceticus–baumannii complex	3 (0·5%)	5 (0·9%)	6 (1·1%)	99·5%	0·25	3	48·5 (41·7–51·1)	2·3 (2·0–4·0)	0·25
Escherichia coli	25 (4·2%)	20 (3·6%)	21 (3·7%)	98·8%	>0·99	17	42 (37·3–58·0)	2·6 (1·9–3·8)	<0·0001
Enterobacter cloacae complex	15 (2·5%)	11 (2·0%)	12 (2·1%)	99·5%	>0·99	10	35·4 (22·9–44·8)	2·0 (1·6–2·4)	0·0020
Haemophilus influenzae	9 (1·5%)	12 (2·1%)	60 (10·7%)	91·5%	<0·0001	12	NA‡	1·9 (1·6–2·6)	NA
Klebsiella aerogenes	8 (1·4%)	8 (1·4%)	6 (1·1%)	99·1%	0·50	6	25·9 (19·3–45·5)	1·7 (1·5–1·9)	0·031
Klebsiella oxytoca	10 (1·7%)	9 (1·6%)	6 (1·1%)	99·1%	0·38	5	28·1 (22·7–41·8)	2·0 (1·7–2·3)	0·062
Klebsiella pneumoniae group	14 (2·4%)	15 (2·7%)	20 (3·6%)	98·8%	0·13	14	37·6 (27·8–44·5)	2·4 (1·8–3·3)	0·0001
Moraxella catarrhalis	2 (0·3%)	2 (0·4%)	12 (2·1%)	98·2%	0·0016	2	41·9 (20·5–63·3)	4·4 (3·1–5·7)	0·50
Proteus species	3 (0·5%)	2 (0·4%)	3 (0·5%)	99·8%	>0·99	2	23·0 (19·7–26·2)	3·1 (2·0–4·1)	0·50
Pseudomonas aeruginosa	38 (6.5%)	34 (6.0%)	39 (6·9%)	98·9%	0·22	33	36·4 (23·1–41·6)	2·2 (1·8–3·5)	<0·0001
Serratia marcescens	16 (2·7%)	14 (2·5%)	14 (2·5%)	99·3%	>0·99	12	31·3 (24·3–43·7)	1·7 (1·6–2·1)	0·0005
Streptococcus pyogenes	0	1 (0·2%)	1 (0·2%)	100·0%	>0·99	1	45·6 (NA)	1·6 (NA)	>0·99
Staphylococcus aureus	84 (14·3%)	84 (14·9%)	93 (16·5%)	95·2%	0·12	75	29·2 (20·0–47·6)	2·0 (1·7–2·7)	<0·0001
Streptococcus agalactiae	3 (0·5%)	2 (0·4%)	29 (5·2%)	95·2%	<0·0001	2	59·0 (21·2–96·8)	4·4 (1·7–7·0)	0·50
Streptococcus pneumoniae	15 (2·5%)	11 (2·0%)	20 (3·6%)	98·2%	0·0016	10	36·6 (24·1–53·2)	2·2 (2·0–3·2)	0·0020
Chlamydia pneumoniae	NA	NA	0	NA	NA	NA	NA	NA	NA
Mycoplasma pneumoniae	NA	NA	0	NA	NA	NA	NA	NA	NA
Legionella pneumophila	1 (0·2%)	2 (0·4%)	5 (0·9%)	99·3%	0·12	1	NA‡	2·6 (2·4–2·8)	NA

Data are n (%) or median (IQR), unless otherwise indicated. Concordance is defined as either the presence or absence of an on-panel pathogen on the BioFire FilmArray pneumonia panel, conventional culture, and antimicrobial susceptibility testing within the intervention group. NA=not applicable. *McNemar test. †Wilcoxon signed rank test. ‡No growth detected in the control group in concordant samples.

Table 2: **Comparison of microorganism detection between the BioFire FilmArray pneumonia panel and conventional culture**

367 (65·2%) participants in the intervention group and 403 (68·4%) in the control group had at least one antimicrobial modification (p=0·24; figure 2A).

For all antibiotics, median time to first antibiotic modification was 20·4 h (95% CI 18·0–20·4) in the intervention group and 25·8 h (22·0–28·7) in the control group (p=0·076; figure 2A). Any escalation of antibiotics against Gram-positive or Gram-negative bacteria was reported in 136 (24·2%) participants in the intervention group and 161 (27·3%) in the control group (p=0·21), with a median time to escalation of 13·8 h (95% CI 9·2–19·0) in the intervention group and 24·1 h (19·5–29·6) in the control group (p=0·0022; figure 2A). Furthermore, escalation of antibiotics against Gram-positive bacteria occurred in 55 (9·8%) in the intervention group and 89 (15·2%) in the control group (p=0·0057), with a median time to change of 10·3 h (95% CI 6·2–30·9) in the intervention versus 24·6 h (19·5–37·2) the control group (p=0·044). The occurrence of escalation against Gram-negative bacteria was similar between both groups (115 [20·5%] of 563 in the intervention *vs* 130 [22·1%] of 589 the control group; p=0·50); however, time to first antibiotic modification occurred at 17·3 h (95% CI 10·8–23·3) in the intervention and 27·2 h (21·3–33·9) in the control group (p=0·010). Overall, any antibiotic de-escalation was reported in 306 (54·4%) participants in the intervention and 340 (57·7%) in the control group (p=0·25), with the median time to first antibiotic de-escalation occurring at similar times in both arms (p=0·37; figure 2A). De-escalation of antibiotics against Gram-positive bacteria occurred in 158 (29·7%) of 532 participants with available data in the intervention and 198 (34·9%) of 567 participants with available data in the control group (p=0·065), with a shorter median time to first antibiotic modification in the intervention than the control group (20·7 h [17·8–24·0] *vs* 27·8 h [22·9–33·0]; p=0·015). De-escalation of antibiotics with Gram-negative spectrum occurred in 231 (42·5%) of 544 participants with available data in the intervention and in 236 (40·8%) of 579 in the control group (p=0·56), with a similar time to first antibiotic modification in both groups (p=0·46; figure 2A).

Among the 514 participants in the pneumonia cohort, any antibiotic modification was reported in 184 (72·4%) of 254 participants in the intervention group and 211 (81·2%) of 260 participants in the control group (p=0·019), with a similar median time to first antibiotic modification in both groups (p=0·46; figure 2B). The probability of escalation of antibiotics against Gram-positive bacteria was reported in 28 (11·0%) of 254 participants in the intervention group versus 55 (21·2%) of 259 participants with available data in the control group (p=0·0017), with a similar time to first antibiotic modification in both groups (p=0·38). However, although median time to de-escalation for antibiotics against Gram-positive bacteria was faster in the intervention than the control group (20·8 h [95% CI 17·0–31·1] *vs* 29·7 h [25·1–34·0]; p=0·043), the occurrence of de-escalation of antibiotics against Gram-positive bacteria was higher in the

	Intervention group (n=563)	Control group (n=589)	p value
None	44 (7·8%)	51 (8·7%)	0·60
Vancomycin	233/519 (44·9%)	271/538 (50·4%)	0·075
Cefepime	195/519 (37·6%)	210/538 (39·0%)	0·62
Piperacillin–tazobactam	163/519 (31·4%)	185/538 (34·4%)	0·30
Ceftriaxone	159/519 (30·6%)	143/538 (26·6%)	0·14
Azithromycin	80/519 (15·4%)	102/538 (19·0%)	0·13
Trimethoprim–sulfamethoxazole	86/519 (16·6%)	96/538 (17·8%)	0·58
Metronidazole	54/519 (10·4%)	63/538 (11·7%)	0·50
Doxycycline	66/519 (12·7%)	59/538 (11·0%)	0·38
Meropenem	34/519 (6·6%)	54/538 (10·0%)	0·040
Cefazolin	62/519 (11·9%)	48/538 (8·9%)	0·11
Levofloxacin	51/519 (9·8%)	45/538 (8·4%)	0·41
Cefdinir	22/519 (4·2%)	27/538 (5·0%)	0·55
Amoxicillin–clavulanate	22/519 (4·2%)	19/538 (3·5%)	0·55
Ertapenem	6/519 (1·2%)	8/538 (1·5%)	0·64

Data are n (%) or n/N (%). Antibiotics listed in order of more commonly prescribed.

Table 3: **Antibiotics administered within 96 h of randomisation**

control group (76 [31·3%] of 153 *vs* 110 [44·0%] of 250; p=0·0036). No differences in escalation (70 [27·7%] of 253 *vs* 79 [30·4%] of 260; p=0·50) or de-escalation (108 [44·3%] of 244 *vs* 116 [45·7%] of 254; p=0·75) of antibiotics against Gram-negative bacteria were observed between the study groups, nor in median time to first escalation or de-escalation against Gram-negative bacteria (figure 2B).

Overall, clinical outcomes were similar between both study groups (table 4). Among participants in the pneumonia cohort not in the ICU at randomisation, ICU admission within 96 h after randomisation was lower in the intervention than in the control group (8·9% *vs* 23·0%; p=0·014). Of 117 deaths in the cohort of participants with pneumonia, 40 (34·2%) were considered to be related to pneumonia, of which 19 (29·2%) of 65 occurred in the intervention group and 21 (40·4%) of 52 in the control group (p=0·23).

ASP recommendations to PCCTs occurred for 121 (21·5%) of 563 participants in the intervention group and for 87 (14·8%) of 589 in the control group (p=0·0034) and more often within the first 24 h in the intervention group than in the control group (68 [56·2%] of 121 participants *vs* 27 [31·0%] of 87). In the control group, ASP interventions occurred mostly between 24 h and 48 h (42 participants [48·3%]). The most common recommendation was to discontinue one or more antibiotics (60 [49·5%] in the intervention group *vs* 49 [56·3%] in the control group), followed by de-escalation of therapy (42 [34·7%] *vs* 37 [42·5%]). ASP recommendations included escalation (13 [10·7%] *vs* four [4·6%]), shortening, or defining durations of antibiotic therapy; dose optimisation; or suggesting infectious disease consultation.

For the study cohort, 30-day acquisition of multidrug-resistant organisms with or without hospital-acquired infection or infection with *C difficile* were not significantly different between groups. Ventilator-free days were unknown due to an error in data collection. Information on antibiotic stops, whether continued beyond 96 h of randomisation or not, was collected as binary (yes or no); therefore, durations of antibiotic therapy were not calculated.

Time to first antibiotic modification was similar between groups, when comparing participants with no pathogen detected by the BioFire FilmArray pneumonia panel in the intervention group and participants with usual flora or no growth in the control group (appendix pp 29–30). Time to first antibiotic modification was also similar between groups when considering: participants with a single positive identification of *S aureus, P aeruginosa, S pneumoniae, E coli,* or *S marcescens* by the panel in the intervention group and by culture in the control group (appendix pp 30–31); participants with panel identification of *S agalactiae, M catarrhalis,* or *H influenzae* in the intervention group and participants with usual flora in the control group (appendix pp 32–33); participants with hospital-acquired pneumonia and those with ventilator-associated pneumonia (appendix pp 33–34); participants with no pathogen identification in either group (appendix pp 34–35); participants with COVID-19 (appendix pp 35–37); and participants without any infection (appendix pp 37–38). No differences in time to first antibiotic modification were observed when comparing between expectorated sputum, bronchoalveolar lavage, and tracheal secretions among all participants of the pneumonia cohort in the intervention group (appendix p 38). The probability of broadening antibiotics against Gram-positive organisms (13 [14·6%] of 89 participants in the intervention group *vs* 18 [7·3%] of 248 in the control group; p=0·042) or Gram-negative organisms 31 [34·8%] of 89 *vs* 30 [12·1%] of 248; p<0·0001) was higher when considering participants with positive detection of *S agalactiae, M catarrhalis,* or *H influenzae* on the BioFire FilmArray pneumonia panel in the intervention group versus participants with usual flora only in the control group.

Discussion

To our knowledge, this study is the largest randomised controlled trial that has evaluated the use of the BioFire FilmArray pneumonia panel with concurrent antibiotic stewardship for patients with suspected pneumonia in a real-world setting. Although many studies show that NAATs provide timely and increased detection of on-panel pathogens from respiratory specimens,[7,9,10,13–15,22] to date there is scarce evidence of a clear effect on antimicrobial use or clinical outcomes. This study showed that, overall, time to any antibiotic modification was not affected by use of the panel. Some delays in antibiotic modification in the intervention group might reflect uncertainty regarding a new test, anticipation of pending conventional culture and AST results, or results of another standard-of-care evaluation, such as blood cultures or plans to proceed to bronchoscopy for bronchoalveolar lavage fluid collection for testing. However, when analysed specifically by antibiotics directed at Gram-positive or Gram-negative bacteria, use of the panel was associated with faster time to escalation of antibiotics overall and a de-escalation of antibiotics against

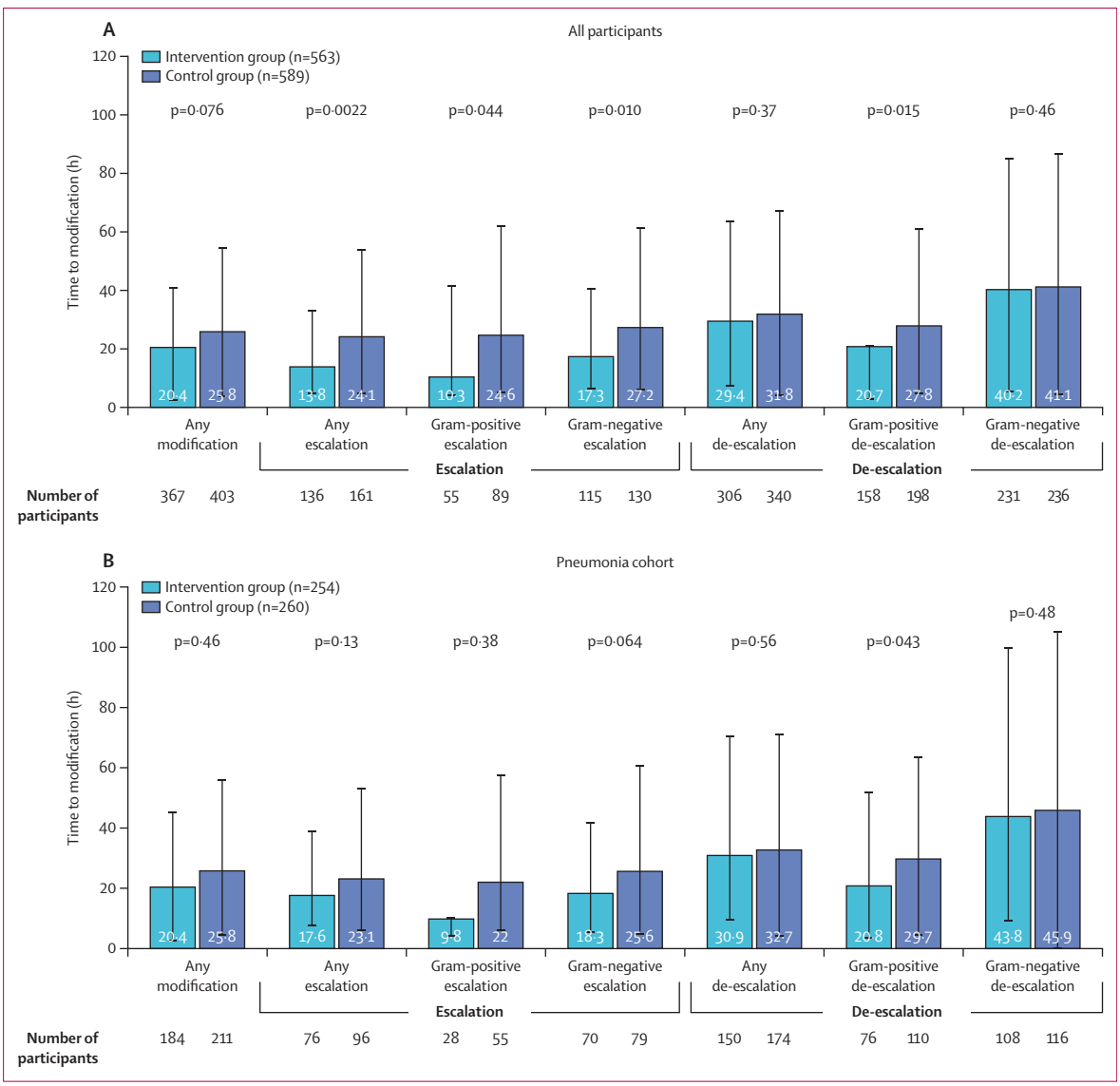


Figure 2: **Median time to first antibiotic modification by study group**
Time to modification (ie, escalation and de-escalation, including for antibiotics against Gram-negative and Gram-positive bacteria) stratified by the cohort including the modified intention to treat population (A) and the pneumonia cohort (ie, participants with community-acquired, hospital-acquired, or ventilator-associated pneumonia; B). Error bars represent 95% CIs.

Gram-positive bacteria in the modified intention-to-treat population and in the pneumonia cohort. Additionally, the probability of escalation of antibiotics against Gram-positive bacteria was higher in the control group than in the intervention group. De-escalation of antibiotics against Gram-negative bacteria was not significantly affected by use of the panel. Use of the panel was associated with more frequent and faster ASP recommendations to PCCTs than with conventional culture and AST alone. The most common recommendations were to discontinue at least one antibiotic or narrow the spectrum. Overall, these results show that use of the BioFire FilmArray pneumonia panel can be associated with faster antibiotic modifications than standard of care in patients being evaluated for pneumonia when focusing on escalation of antibiotics directed to Gram-positive or Gram-negative pathogens or de-escalation of antibiotics directed to Gram-positive pathogens. The study also showed a potential for increased antibiotic escalation, appropriate or not, in patients in whom *S agalactiae*, *M catarrhalis*, or *H influenzae* was detected by the panel compared with those with reported usual flora in the control group.

Although improved clinical outcomes following pneumonia, particularly mortality, have been reported with timely (<4 h) pathogen-appropriate antibiotic therapy,[23] this study showed that, despite faster pathogen detection with

	Modified ITT cohort			Pneumonia cohort		
	Intervention group (n=563)	Control group (n=589)	p value	Intervention group (n=254)	Control group (n=260)	p value
Death within 30 days of randomisation	127/563 (22·6%)	120/589 (20·4%)	0·42*	65/254 (25·6%)	52/260 (20·0%)	0·20*
Length of stay in hospital, days	9·0 (3·0–18·0)	8·0 (4·0–19·0)	0·42†	9·0 (4·0–19·0)	9·0 (5·0–19·0)	0·66†
Length of stay in ICU‡, days	6·5 (3·0–14·0)	6·0 (3·0– 14·0)	0·72	7·0 (3·0–18·0)	6·0 (3·0–13·0)	0·19
Re-admission within 30 days of randomisation	79/563 (14·0%)	102/589 (17·3%)	0·30*	33/254 (13·0%)	44/260 (16·9%)	0·39*
Admission to ICU at 96 h§	18/206 (8·7%)	28/219 (12·8%)	0·18*	7/79 (8·9%)	20/87 (23·0%)	0·014*
Admission to ICU at 30 days§	18/206 (8·7%)	17/219 (7·8%)	0·71*	5/79 (6·3%)	9/87 (10·3%)	0·35*
Hospital-onset *Clostridioides difficile* infection	10/563 (1·8%)	15/589 (2·5%)	0·37*	8/254 (3·1%)	7/260 (2·7%)	0·76*
Acquisition of multidrug-resistant microorganisms with or without hospital-acquired infection within 30 days of randomisation	44/370 (11·9%)	60/395 (15·2%)	0·205	27/181 (14·9%)	27/191 (14·1%)	0·88

Data are n (%) or median (IQR), unless otherwise indicated. The modified ITT population included all participants whose specimen met all eligibility criteria and no exclusion criteria. The pneumonia cohort included all participants with community-acquired, hospital-acquired, or ventilator-associated pneumonia. Antibiotic duration was not calculated because antibiotic continuation at 96 h was only collected as a binary yes or no response. ITT=intention-to-treat. ICU=intensive care unit. *Chi squared test. †Wilcoxon rank-sum test. ‡For patients in ICU at randomisation or admitted within 24 h of randomisation. §Not in ICU at randomisation.

Table 4: **Secondary clinical outcomes**

use of the BioFire FilmArray pneumonia panel, there was no effect on most clinical outcomes, including mortality. However, fewer admissions to the ICU were observed among participants with pneumonia in the intervention group than in the control group.

Antimicrobial de-escalation is important to decrease selective antibiotic pressure. Although de-escalation of antibiotics against Gram-positive bacteria was substantiated by confidence in the BioFire FilmArray pneumonia panel, de-escalation of antibiotics with Gram-negative spectrum was disappointing despite the observed rapidity and reliability of the panel. This finding is consistent with other data showing that microbiological results in patients with pneumonia are often not acted on for de-escalation,[3,24,25] suggesting that de-escalation of antibiotics against Gram-negative bacteria, in particular, requires more complex decision making with higher trust in the validity of test results to allay clinician fears of negative consequences, especially in severely ill patients. Pneumonia practice guidelines[1] recommend de-escalation of broader spectrum antibiotics, including anti-pseudomonal antibiotics, after negative high-quality microbiological tests, irrespective of invasively or non-invasively obtained samples. However, recommendations from the Infectious Diseases Society of America[1] to de-escalate anti-pseudomonal antibiotics in the absence of pseudomonal growth are based on retrospective studies. A US study showed that antimicrobial de-escalation was performed in less than 30% of patients admitted for hospital-acquired pneumonia between 2008 and 2011.[26] Among the 35% of participants who had a sputum culture obtained, de-escalation occurred more commonly in those with positive culture results than in those with negative results. The study also showed that, despite the absence of MRSA or *P aeruginosa* in sputum, therapy directed at these pathogens was frequently continued.[26] De-escalation was more common in patients with positive blood cultures than those with a positive respiratory culture only, suggesting a higher degree of confidence in blood cultures for management of pneumonia.[26] Bacteraemia occurs infrequently in patients with pneumonia (41 [4·3%] of 963 participants in our study). This suggests that tests with accurate and broad pathogen detection, such as the BioFire FilmArray pneumonia panel, should help with de-escalation. Compared with the broader adoption of vancomycin de-escalation based on evidence of the high negative predictive value of negative MRSA nasal screening for MRSA pneumonia,[27] similar high-quality studies are scarce for de-escalation of antibiotics targeted at Gram-negative bacteria (albeit generally advised). A 2022 study of patients with hospital-acquired pneumonia showed that use of the Unyvero NAAT pneumonia panel (Curetis, Holzgerlingen, Germany) facilitated de-escalation of antibiotics directed at Gram-negative pathogens in 61 (66%) of 92 participants.[18] There are several potential reasons for the low rates of de-escalation of antibiotics against Gram-negative bacteria observed here, including high severity of illness (more than 60% of participants were in the ICU at randomisation with a median overall Charlson Comorbidity Index score of 4), limited evidence of a strong negative predictive value of a negative BioFire FilmArray pneumonia panel, and overlapping clinical findings between infectious and non-infectious respiratory conditions. Although antimicrobial stewardship in patients with pneumonia can improve antimicrobial use,[28,29] this study highlights the need to better define safe de-escalation strategies. Although use of the BioFire FilmArray pneumonia panel improves the diagnostic yield for patients with pneumonia, there is a risk of inappropriate antibiotic use with detection of commensal or colonising bacteria. This risk suggests additional diagnostic and antimicrobial stewardship efforts are needed to realise the most appropriate benefit of the BioFire FilmArray pneumonia panel. Not discontinuing antibiotics early in patients without pneumonia might be due to the inclusion of patients with non-pulmonary infections, severe immunocompromise, or undefined pulmonary syndromes.

The BioFire FilmArray pneumonia panel provides an advance in pneumonia diagnostics; however, because it detects only some pathogens and antibiotic resistance

determinants, it cannot replace standard of care (ie, conventional culture and AST). When performed, the panel could potentially replace some adjunctive testing, such as nasal MRSA screening, standalone molecular assays for pathogens, or urine pneumococcal or *Legionella* antigen tests.

Strengths of this study include the large study population, the randomised controlled trial design, the incorporation of antibacterial guidance via EHR comments, and ASP recommendations in a real-world setting. Including several ASP physicians and pharmacists allowed robust assessment of the effect of the BioFire FilmArray pneumonia panel. An additional arm without antimicrobial stewardship would have allowed for an evaluation of the effect of antibiotic modifications solely by PCCTs based on EHR guidance comments. The study workflow is applicable and generalisable to other institutions that adopt the BioFire FilmArray pneumonia panel and have an ASP.

Limitations of this study include that respiratory testing was obtained for various reasons and at various points of clinical care, with some testing performed for unclear reasons; more than half of participants ultimately had diagnoses other than pneumonia. This limitation confounds the accuracy of the primary outcome of antibiotic modifications and assessment of whether modifications were related to specific test results. Evaluating appropriateness of antibiotics for specific pathogens in patients with community-acquired, hospital-acquired, or ventilator-associated pneumonia and according to sample type (ie, sputum, tracheal secretions, or bronchoalveolar lavage) should be a focus of future research. Another limitation of the study is that the total duration of antibiotic therapy was not assessed because the effect of the BioFire FilmArray pneumonia panel was expected be immediate and within the window of pending availability of conventional culture and AST (72–96 h). Use of a shorter course of antibiotics for pneumonia mighty affect decisions to modify antibiotics, making it harder to understand whether timing of the test in relation to presentation affects time to first antibiotic modification (ie, inertia to change antibiotics if only 1 or 2 days of total course are left). The role of adjunctive nasal MRSA screening in the time to first antibiotic modification is unclear; however, the number of participants who underwent this test or tested negative was not significantly different between both groups. Multiplicity of ASP staff and absence during off hours, weekends, or holidays might have affected antibiotic adjustments. However, guidance comments added to the microbiology report were surrogates of similar microbe-directed information for when ASP members were unavailable. The specific role that the guidance comments had in antibiotic modification was unassessed and could be a future area of research, potentially expanded on by the incorporation of artificial intelligence. Furthermore, the low detected rates of antibiotic resistance determinants in Gram-negative microorganisms might limit generalisation to institutions with high resistance rates. Although this study shows that the BioFire FilmArray pneumonia panel could assist in faster antibiotic modifications based on the rapidity of results, future studies should evaluate strategies that enable improved test use and antibiotic de-escalation.

In conclusion, the BioFire FilmArray pneumonia panel has a potential role in the management of patients with suspected pneumonia because its use was associated with a shorter time to escalation of antibiotics directed at Gram-positive and Gram-negative bacteria and to de-escalation of antibiotics directed at Gram-positive bacteria compared with standard testing based on conventional culture and AST. Use of the BioFire FilmArray pneumonia panel was also associated with more rapid interventions by the ASP team and decreased admission to the ICU among patients with pneumonia.

Contributors

AV, RP, and JM contributed to the study design. WSH and JM did the statistical analysis. AV wrote the first draft of the manuscript, with input from WSH, JM, and RP. LMH, KAR, APS, KDK, and ADD assisted in recruitment, randomisation, and inter-team communication. AV, ANS, RP, RWS, LLE, AJT, DWC, DRO, MF, PV, JWW, GAS, and NSR participated in the study procedures. RD and BJB assisted in the cost analysis. AV, WSH, and JM accessed and verified the underlying data reported in the manuscript. All authors assume responsibility for the accuracy and completeness of the data and analyses. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

RP and AV received funding from bioMérieux for this study. AV also received funding from Moderna for advisory board participation in 2021. AV reports being an inventor for Mayo Clinic Travel App interaction with Smart Medical Kit and Medical Kit for Pilgrims. PV reports research funding from Cidara, Scynexis, and Ansun; and serves as a consultant for AbbVie and Scynexis (all fees paid to Mayo Clinic). PV reports honoraria from the Merck Manuals and *Current Fungal Infection Reports*. RP reports grants from ContraFect, TenNor Therapeutics, and bioMérieux. RP is a consultant to PHAST, Torus Biosystems, Day Zero Diagnostics, Mammoth Biosciences, and HealthTrackRx; monies are paid to Mayo Clinic. Mayo Clinic and RP have a relationship with Pathogenomix. RP has research supported by Adaptive Phage Therapeutics. Mayo Clinic has a royalty-bearing, know-how agreement and equity in Adaptive Phage Therapeutics. RP is also a consultant to Netflix, Abbott Laboratories, Oxford Nanopore Technologies, and CARB-X. RP has an issued patent on *Bordetella pertussis–parapertussis* PCR; a patent issued on a device for sonication, with royalties paid by Samsung to Mayo Clinic; and an issued patent on an anti-biofilm substance. RP receives honoraria from the National Board of Medical Examiners, Up-to-Date, and the Infectious Diseases Board Review Course. AJT receives honoraria from Up-To-Date. All other authors declare no competing interests.

Data sharing

The study's methods and actions are published in the appendix (pp 1–39). De-identified participant data can be made available if requested to the corresponding author after publication; however, this will be contingent on the investigator's review of the request details, approval from the investigator's institutional research board, and the completion of a signed data access agreement.

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